COMP4141 Slides Week 8

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Admin

Feedback

For the star system:

1. Minimally satisfactory: has a construction that can be made to work, shows some understanding of the proof, but it could have been done in more detail.
2. correct construction with a good level of proof details, only minor errors
3. Excellent answer with very detailed and correct proof.

Again stars do **not** count towards marks.

P and NP

Definition of P and NP

Definition (P)

$$\mathsf{P} = \bigcup_{k \in \mathbb{N}} \mathsf{TIME}(n^k),$$

i.e. the class of problems that can be solved in polynomial time.

Definition (NP)

$$\mathsf{NP} = \bigcup_{k \in \mathbb{N}} \mathsf{NTIME}(n^k),$$

i.e. the class of problems that can be solved in nondeterministic polynomial time.

Verifier Characterisation of NP

Verifier Definition

A verifier for a language A is an algorithm V such that

$$A = \{ w \mid V \text{ accepts } \langle w, c \rangle \text{ for some string } c \}.$$

The language A is polynomially verifiable if it has a polynomial-time verifier (where the running time is measured in terms of $|w|$).

Theorem

A language is in NP if and only if it is polynomially verifiable.

Polynomial-Time Reductions

Definition (Polynomial-Time Computable)

A function $f : \Sigma^* \to \Sigma^*$ is polynomial-time computable if there exists a polynomial-time TM M that, on input $w \in \Sigma^*$, halts with $f(w)$ on its tape.

Definition (Polynomial-Time Mapping Reduction)

Let $A \subseteq \Sigma_1^*$ and $B \subseteq \Sigma_2^*$. We say $A \leq_P B$ if there exists a polynomial-time computable function $f : \Sigma_1^* \to \Sigma_2^*$ such that $w \in A \iff f(w) \in B$.

NP-hard

Definition

A language B is NP-hard if every $A \in \mathrm{NP}$ satisfies $A \leq_P B$.

Theorem (Proving NP-hardness)

If B is NP-hard and $B \leq_P C$, then C is NP-hard.

Typical method: reduce a known NP-hard problem P_1 to a new problem P_2. This is easier said than done!

NP-completeness

Definition

A language B is NP-complete if:

1. $B \in \mathrm{NP}$,
2. B is NP-hard.

SAT

Definition

Let Prop $= \{x, y, \ldots\}$ be Boolean variables. A Boolean formula is defined inductively by:

$$\varphi \to p \mid \varphi \wedge \varphi \mid \neg\varphi \mid (\varphi), \quad p \to x \mid y \mid \cdots$$

Its truth depends on assignments to the variables.

SAT

Input: A Boolean formula φ

Question: Is φ satisfiable?

The Grandfather of NP-Complete Problems

Theorem

SAT is NP-complete.

The proof uses a general reduction for hardness and a satisfying assignment as a certificate to show membership in NP.

3SAT

Definition

3-CNF

Input: A Boolean formula φ

Question: Is φ satisfiable?

A formula is in 3-CNF if it is a conjunction of clauses, where each clause has at most three literals, e.g.

$$(x \vee y \vee z) \wedge (x \vee y \vee \neg z) \wedge (x \vee \neg y \vee z).$$

W8 P1

Problem 1

Say that a formula is 4-CNF if it is in conjunctive normal form, with each clause having at most 4 literals.

4-SAT

Input: A 4-CNF formula ϕ

Question: Is ϕ satisfiable?

Show that 4-SAT is NP-Complete.

W8 P2

Problem 2

Consider the following decision problems:

Hamiltonian Path

Input: An undirected graph G

Question: Is there a path in G that visits every vertex exactly once?

Hamiltonian Cycle

Input: An undirected graph G

Question: Is there a cycle in G that contains every vertex exactly once?

(a) Show that Hamiltonian Path $\leq_P$ Hamiltonian Cycle.

Problem 2

Consider the following decision problems:

Hamiltonian Path

Input: An undirected graph G

Question: Is there a path in G that visits every vertex exactly once?

Hamiltonian Cycle

Input: An undirected graph G

Question: Is there a cycle in G that contains every vertex exactly once?

(b) Show that both problems are in **NP**.

W8 P3

Cliques and Vertex Covers

Definition

A *clique* in an undirected graph is a fully connected subgraph. A k-clique is a clique with k vertices.

Vertex Cover

A *vertex cover* in an undirected graph $G = (V, E)$ is a set $\mathcal{V}$ such that for every edge $(u, v) \in E$, at least one of u or v belongs to $\mathcal{V}$.

Problem 3

Consider the following problems:

Clique

Input: An undirected graph G and an integer k

Question: Does G have a clique of size k?

Vertex Cover

Input: An undirected graph G and an integer k

Question: Does G have a vertex cover of size k?

Show that Clique is **NP**-complete by giving a reduction from Vertex Cover.

W8 P4

Problem 4

Consider the following problem:

3-Col

Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

Suppose we knew that 3-Col is **NP**-complete. Using this fact, give a new proof by reduction that SAT is **NP**-complete.

W8 P5

Problem 5

Consider the following problem:

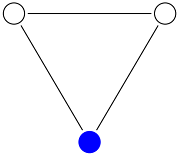
3-Col

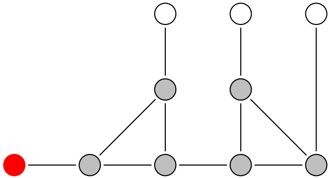
Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

Problem 5

(a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?





Problem 5

(b) Show that 3-Col is **NP**-complete, assuming that SAT is **NP**-complete.

W8 P6

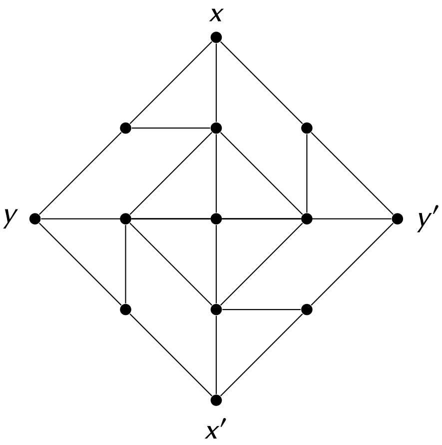
Problem 6

Show that if $\mathbf{P} = \mathbf{NP}$, then every language $A \in \mathbf{P}$ except $A = \emptyset$ and $A = \Sigma^*$, is **NP**-complete.

Bonus

Bonus

Consider the following graph:



Bonus

(a) Give a valid 3-colouring of this graph where x and y have different colours.

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(b) Give a valid 3-colouring of this graph where x and y have the same colour.

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(b) Give a valid 3-colouring of this graph where x and y have the same colour.

(c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

Bonus

(a) Give a valid 3-colouring of this graph where x and y have different colours.

(b) Give a valid 3-colouring of this graph where x and y have the same colour.

(c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

(d) Consider the following decision problem:

Planar 3-Col

Input: A planar graph G

Question: Does G have a valid 3-colouring?

Show that Planar 3-Col is NP-complete.
Hint: Use the above graph to "uncross" edges.

Bonus

(e) Prove that there is no planar graph G, with x, y, x', y' on an outer face (in that order), that satisfies the conditions indicated in (a), (b) and (c), but with a 4-colouring rather than a 3-colouring.